

# altairsim — examples

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**Machines that boot.** Each directory here is self-contained: a `.toml` that describes the machine, the media it needs lying beside it, and a note saying what you will see. Copy any one of them anywhere and it still runs — a path inside a machine file resolves against **that file**, not against the directory you launched from.

```
altairsim examples/cpm/cpm22-buffered.toml    # CP/M 2.2b on an 8" floppy
altairsim examples/cpm/cpm22-terminal.toml    # ...the same, console in a built-in VT100 window
altairsim examples/basic/basic4k.toml         # Altair 4K BASIC, off a 1975 cassette
altairsim examples/diskbasic/diskbasic.toml   # Altair Disk Extended BASIC 4.1 on an 8" floppy
altairsim examples/uio/uio.toml              # Altair 8K BASIC over ONE 88-UIO (serial +
cassette)
altairsim examples/tarbell/tarbell.toml       # CP/M that boots itself from a Tarbell PROM
altairsim examples/hdsk/hdsk.toml            # CP/M off an 88-HDSK hard disk
altairsim examples/turnkey/floppy.toml        # an 8800bt Turnkey Module, no front panel
altairsim examples/cdos/cdos.toml            # Cromemco CDOS 2.58 off a 16FDC
altairsim examples/sdsys/sbc200.toml         # the SD Systems SBC-200 Z80 single-board computer
altairsim examples/sol/trek80.toml           # a Sol-20 at SOL05, with TREK80 in the deck
altairsim examples/debugger/debugger.toml    # a bench for learning the symbolic debugger
altairsim examples/dazzler/kscope.toml       # a Cromemco Dazzler, drawing a kaleidoscope
altairsim examples/printing/printer.toml     # an 88-C700 printer, wired to a real host printer
```

|                         | What it is                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>cpm/</code>       | Mike Douglas's track-buffered <b>CP/M 2.2b v2.3</b> , 56K, booted by the DBL PROM from an 8" floppy. <code>A&gt;</code> in one command. <code>cpm22-terminal.toml</code> boots the same disk with its console in a <b>built-in VT100 window</b> the simulator draws itself — no telnet client, no external emulator.                                                                                                                         |
| <code>basic/</code>     | <b>Altair 4K BASIC 3.1</b> ( <code>basic4k.toml</code> ) read off a period <code>.tap</code> by the bootstrap MITS shipped, unmodified: <code>MEMORY SIZE?</code> . Beside it, <b>8080 BASIC VER 1.0</b> ( <code>basic1.toml</code> ) — the first Altair BASIC, 1975 — booted the raw two-step way its primitive bootstrap forced: <code>RUN 1800</code> , <code>^E</code> , <code>RUN 0</code> → <code>MEMSIZ?</code> .                     |
| <code>diskbasic/</code> | <b>Altair Disk Extended BASIC 4.1</b> (MITS, 1977) on an 8" floppy behind an 88-DCDD, booted by the DBL PROM. Unlike the cassette BASIC, this one has a filesystem — <code>SAVE</code> by name, a directory, <code>DSKINI</code> . <code>MEMORY SIZE?</code>                                                                                                                                                                                 |
| <code>uio/</code>       | <b>Altair 8K BASIC 3.2</b> over a single <b>88-UIO</b> — the board that is a 6850 serial port (at 0x10) <i>and</i> an 88-ACR cassette (at 0x06) in one. The period bootstrap runs unmodified. <code>MEMORY SIZE?</code>                                                                                                                                                                                                                      |
| <code>tarbell/</code>   | The <b>Tarbell</b> single- and double-density floppy interfaces, which carry their own 32-byte boot PROM: power on with a disk in the drive and 48K <b>CP/M 2.2</b> comes up — no monitor, nothing to type.                                                                                                                                                                                                                                  |
| <code>hdsk/</code>      | <b>CP/M 2.2</b> booting off an Altair <b>88-HDSK "Datakeeper"</b> hard disk — an outboard controller moving whole sectors over an 88-4PIO — loaded by the HDBL PROM. <code>A0&gt;</code>                                                                                                                                                                                                                                                     |
| <code>turnkey/</code>   | The <b>MITS 8800bt Turnkey Module</b> : an Altair with no front panel that boots itself the moment it powers on. Two files boot CP/M two ways — off an 88-DCDD floppy and off an 88-HDSK hard disk.                                                                                                                                                                                                                                          |
| <code>cdos/</code>      | <b>Cromemco CDOS 2.58</b> , Cromemco's CP/M work-alike, booted from an 8" diskette through a <b>Cromemco 16FDC</b> — the board carrying the disk controller, console UART, and boot PROM at once.                                                                                                                                                                                                                                            |
| <code>sdsys/</code>     | The <b>SD Systems SBC-200</b> , a whole Z80 computer on one S-100 board (8251 console, Z80-CTC, boot PROMs) running the <b>MSMONR21</b> monitor; with a VersaFloppy it boots SDOS or CP/M.                                                                                                                                                                                                                                                   |
| <code>sol/</code>       | A <b>Processor Technology Sol-20</b> running SOLOS 1.3, with the 1977 game <b>TREK80</b> on cassette. Type <code>XE TRK80</code> .                                                                                                                                                                                                                                                                                                           |
| <code>debugger/</code>  | A 46-byte program with its <b>symbols</b> and a guided walk through the monitor's debugger: <code>SYMBOLS LOAD</code> , symbolic <code>DISASM</code> , single-step, break on a label, run.                                                                                                                                                                                                                                                   |
| <code>dazzler/</code>   | A <b>Cromemco Dazzler</b> , the S-100's first color graphics card, running Li-Chen Wang's <b>Kaleidoscope</b> . Comes up drawing a four-way-mirrored pattern in a window; <code>ATTN</code> (Ctrl-E) breaks back to the monitor.                                                                                                                                                                                                             |
| <code>pmmi/</code>      | A <b>PMMI MM-103 modem</b> and a small 8080 <b>terminal</b> program that bridges the console to the phone line. Four control keys work the modem — <code>^D</code> DTR, <code>^S</code> the 6860 Self Test loopback, <code>^I</code> modem status, <code>^C</code> quit. <code>^D</code> then <code>^S</code> makes the card echo your keystrokes with no phone line attached; set <code>dial / answer</code> to place a real call over TCP. |

|                        | What it is                                                                                                                                                                                                                                                                                         |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>printing/</code> | An <b>88-C700 line printer</b> , and a banner program that prints through it. The README sets up a real printer on your host — a network printer over <code>socket:</code> , or a CUPS queue over <code>printer:</code> — and a page comes out. Per-OS host setup (macOS, Linux; Windows pending). |
| <code>ai-mcp/</code>   | A working directory for an <b>AI assistant driving altairsim over MCP</b> : a CP/M machine and a tiny <code>HELLO.ASM</code> with one deliberate bug the assistant assembles, runs, single-steps to find, and fixes — all through the simulator's MCP tools. See <code>DRIVING-WITH-AI.md</code> . |

**This tree is the product**, which is the reason it exists as one directory rather than as media scattered through `disks/` and `tapes/` . It is what `tools/build-package.sh` assembles, it is what `docs/manual/quick-start.md` promises, and `acceptance-examples` boots every one of these out of a scratch directory with no repository in sight — because "does it work here" and "does it work where we hand it to people" turned out to be different questions.

Nothing is duplicated. `disks/` and `tapes/` keep only what does **not** ship: the period `.ASM` listings the boards were built from, download stubs for optional images, and vendor documentation.